

```
1 function [numColorAnomalies, colorStats] = colorAnomalyDetection(img)
2
3 hsvImg = rgb2hsv(img);
4 hChannel = hsvImg(:,:,1);
5 sChannel = hsvImg(:,:,2);
6
7 % "Blue Tube" roughly means: Hue between 0.5-0.7, and high saturation
8 blueMask = (hChannel > 0.5) & (hChannel < 0.7) & (sChannel > 0.4);
9
10 % Clean up the mask to remove tiny blue glares or noise
11 blueMask = bwareaopen(blueMask, 500);
12 blueMask = imfill(blueMask, 'holes');
13
14 % Measure the colored tubes
15 colorStats = regionprops(blueMask, 'BoundingBox', 'Area');
16
17 % Track the number of color anomalies we found by counting the number of
18 % rows in the table.
19 % We can ignore this section after we get this information.
20 numColorAnomalies = length(colorStats);
21
22 end
```